PHOTON COINCIDENCE MEASUREMENTS

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Abstract

This work presents a brief characterization of SPADs and their application in correlation measurements. Measurements using lasers, thermal and pseudothermal light sources were performed to understand the limitations of measuring second order coherence. Initial measurements yield the zero-delay coherence value $g^{(2)}(0)$ for pulsed lasers and map the complete correlation function $g^{(2)}(\tau)$. Finally, a rotating diffuser was implemented to generate dynamic speckle patterns providing a demonstration of speckle decorrelation time (τ_c).

Introduction

Quantum optics classifies light sources based on their intrinsic photon emission statistics, which can be quantified using the second-order correlation function, $g^{(2)}(\tau)$ [1]. This function measures the time-separated probability of detecting two photons with a particular temporal delay τ. The distinct behavior of $g^{(2)}(\tau)$ under different illumination conditions enables the study of the statistical properties of light.

Phase-coherent light is characterized by a constant macroscopic phase and exhibits independent photo detections governed by Poissonian statistics. The arrival of one photon provides no predictive information about the next, resulting in a flat correlation of $g^{(2)}(\tau) = 1$. In contrast, thermal light is formed by the superposition of countless independent waves with random, fluctuating phases, resulting in a Gaussian random field. The bosonic nature of photons in such chaotic fields leads to photon bunching. A phenomenon that manifests with high probabilities of detecting photons close together in time. This broadens the distribution into super-Poissonian statistics ($\sigma^2 > \mu$) and yields a zero-delay correlation peak of $g^{(2)}(0) = 2$.

For broadband thermal sources used in this work, the coherence time is much shorter than the detector timing resolution. However, by scattering a coherent continuous-wave laser through a dynamic medium, such as a rotating diffuser, a "pseudothermal" light source is generated. The resulting dynamic speckle pattern, produced by the rotating diffuser, exhibits intensity fluctuations at accessible time scales. This enables the measurement of speckle intensity decorrelation and its dependence on diffuser motion.

These correlations can be accurately measured using a technique known as time-correlated single-photon counting (TCSPC) which relies on single-photon avalanche diode (SPAD) detectors and time-to-digital converters (TDC). The measurement process begins when an incident photon strikes the active area of a SPAD detector. This triggers an avalanche of charge carriers which in turn creates a macroscopic current pulse [2, 3, 4]. This analog signal is then registered by a TDC and converted into a highly precise digital timestamp. By accumulating millions of these individual photodetection events and cross-referencing their arrival times across multiple SPADs, we can measure correlation of a light source.

Three measurements are presented in this work. First, a $g^{(2)}(0)$ measurement under pulsed light illumination is conducted using two SPAD detectors to extract coincidence counts. The number of coincidences is mathematically related to the zero delay correlation value, and we study its dependence on the coincidence threshold Δt. Second, we measure the full second-order coherence function for both thermal and pulsed sources. Finally, the third measurement demonstrates a proof-of-principle to extract the decorrelation times of the dynamic speckle generated by a pseudothermal light source.

Contents

1 Layout and Calibration

The experimental layout used throught the measurements is presented figure 1. The layout is designed to allow interchanging light sources incident on the SPAD. Optical elements can also be introduced between the source and the SPAD for attenuation, modulation and alignment purposes.

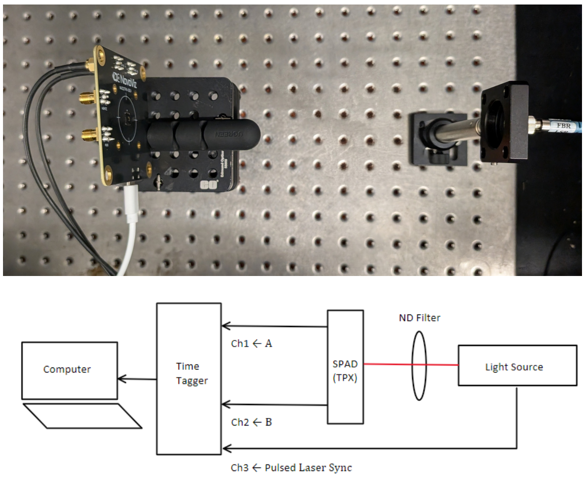

Figure 1: Experimental layout. Source is feed into the system with fiber optic cable that couples into free space with the SM1 FC2 and irradiates the SPAD.

1.1 TDC Application Programming Interface

Voltage pulses generated by the SPAD detectors are time-stamped, and digitalized by a TDC from Swabian Instruments. Furthermore, we utilize the Swabian API [5], and the `tt.Correlation` class for initial digital processing. This function computes the cross-correlation between two detection channels.

The algorithm operates within a user-defined correlation time window. This window is determined by specifying a temporal `bin width` and a total `number of bins`. For every photon event registered on a start channel, the algorithm calculates the time delay (τ) to subsequent events on a stop channel, provided those events fall within the bounded correlation window. These computed delays are continuously accumulated and sorted into their respective time bins resulting in the final coincidence histogram, which is directly proportional to the second-order coherence function, $g^{(2)}(\tau)$.

1.2 SPAD Characterization

With the set up complete, a SPAD calibration is performed to verify the setup is operating within the expected margins. With an autocorrelation measurement we expect to extract a value for the dead time of the detector, and the afterpulsing probability and confirm that the digital pipeline is working as expected. The autocorrelation is measured with the `tt.Correlation` function with the same SPAD as the start and stop channel. The procedure is listed bellow:

1. Connect the CW laser source 1 to the fiber optics line feeding the system.
2. Connect the SPAD detectors with SMA Cables to Swabian Time Tagger.
3. Initialize the autocorrelation measurement (`tt.Correlation`) using a single SPAD channel as both the "start" and "stop" trigger.
4. Wait for detection events to accumulate for 10 seconds.
5. Extract dead time and afterpulsing probabilities from data (See figure 2).
6. Repeat 5 times to gather statistics.

Figure 2 shows the resulting autocorrelation histogram. The dead time is quantitatively defined as the temporal delay at which the coincidence counts recover from 0 counts. To calculate the afterpulsing probability (P_{ap}), the steady-state baseline counts are first estimated from the flat region of the histogram occurring well after the detector recovery. This baseline is subtracted from the total counts to isolate the excess detections. P_{ap} is then determined by integrating these excess counts from the end of the dead time (t_{dt}) to the end of the time window (t_{end}), normalized by the total number of valid trigger events ($N_{triggers}$):

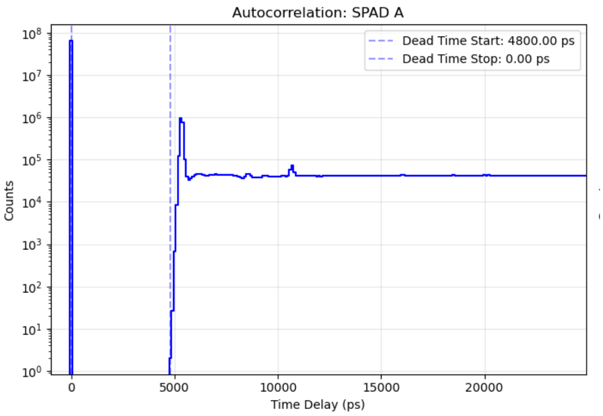

Figure 2: Autocorrelation histogram of the SPAD channel 1. The measurement was taken with an illumination count rate of 10 Mcps and a histogram bin width of 100 ps.

$$P_{ap} = \frac{\sum_{t=t_{dt}}^{t_{end}} (N_{counts}(t) - N_{baseline})}{N_{triggers}} \quad (1)$$

The results of this characterization are summarized in Table 1.

Table 1: Summary of Detector Calibration Parameters

Parameter	SPAD 1	SPAD 2
Mean Dead Time (ns)	4.90 ± 0.12	4.85 ± 0.09
Maximum Repetiton Rate (MHz)	204.08	206.19
Afterpulsing Probability (%)	2.41	2.30

2 Poissonian Statistics Verification

2.1 Coherent Light

This section presents the experimental measurement of photon count distributions, which are used to infer Poissonian and super-Poissonian statistics. Measurements were performed using the CW laser 1 (listed in section 3.3) in the following manner.

- Feed the CW laser 1 into the optical setup in figure 1.
- Mount Filter in between the source and the SPAD.
- Set the integration time window to 100 μs
- Connect SPAD to Swabian Time Tagger.
- Run the `tt.Counter` function and plot histogram.

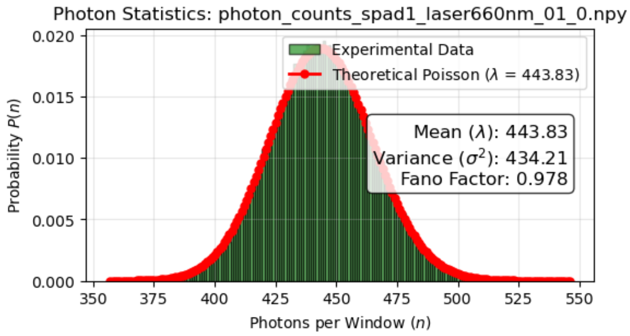

Figure 3: Photon count distribution ($1mW$).

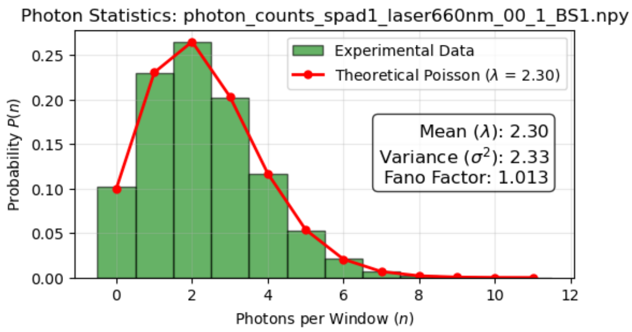

Figure 4: Photon count distribution($0.1mW$).

As shown in Figure 4, the coherent laser source yields a Poissonian distribution even as we vary the power of the CW laser. Furthermore, in Figure 3, a polarizing beam splitter was introduced to measure the curve in the low-power regime, also resulting in a Poissonian distribution. The fact that the variance closely matches the mean ($\sigma^2 \approx \mu$) is consistent with the expected Poissonian statistics of coherent laser light.

2.2 Non-Coherent Light

We then replaced the light source with a fluorescent light bulb, acting as a non-coherent broadband source, and ran the previous experiment with the CW laser turned off. The resulting super-Poissonian distribution is shown in Figure 5.

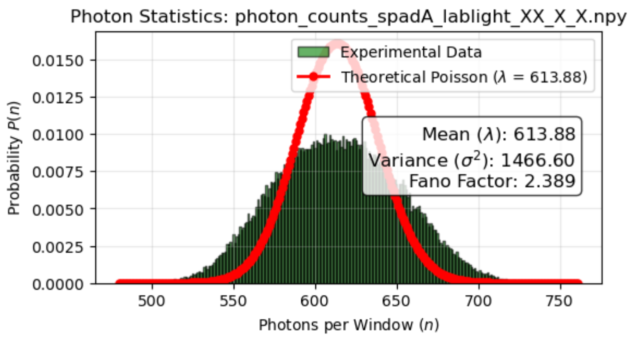

Figure 5: Photon counting statistics of the fluorescent light bulb, utilized here as a non-coherent broadband source. The experimental data follows a super-Poissonian distribution.

3 Coincidence Measurements

3.1 1st Measurement: HBT

The first measurement evaluates the normalized second-order intensity correlation at zero delay, $g^{(2)}(0)$. The procedure is given below:

1. Feed the pulsed laser into the optical setup shown in Figure 1.
2. Select a laser repetition rate within the operational range (1 MHz to 80 MHz), ensuring it remains well below the maximum theoretical rate dictated by the detector dead time.
3. Select the exposure time.
4. Connect the laser sync and SPAD channels to the time tagger.
5. Measure the timestamps of detection events.
6. Compare channel ticks to evaluate coincidences:

$$\Delta t = |t_B - t_A|$$

Coincidence registered if:

$$\Delta t \leq 2 \cdot \text{Jitter}$$

7. Calculate independent detection probabilities:

$$P_A = \frac{\text{Number of Events}_A}{\text{Number of Pulses}}$$

$$P_B = \frac{\text{Number of Events}_B}{\text{Number of Pulses}}$$

$$P_{AB} = \frac{\text{Number of Coincidences}}{\text{Number of Pulses}}$$

8. Calculate the zero-delay correlation:

$$g^{(2)}(0) = \frac{P_{AB}}{P_A \cdot P_B}$$

Unlike a traditional setup, this measurement does not use a beam splitter. Instead, it implements a Hanbury Brown–Twiss (HBT)-type intensity correlation measurement by cross-correlating photon detection events from two neighboring SPAD pixels that simultaneously sample the same optical illumination [6]. This spatial configuration is effective due to the close proximity of the detectors; the pixel pitch is 28 μm for the two-pixel module (NV02TPX). Under ideal coherent pulsed illumination, the independent arrival of photons means the expected theoretical result is $g^{(2)}(0) = 1$.

3.1.1 Pulsed laser Measurements

Measurements of $g^{(2)}(\tau = 0)$ were taken with 10 ps resolution with a laser attenuation adjusted to obtain $\sim$ 2% of total pulses detected by the SPADs. Results are summarized in Figure 6.

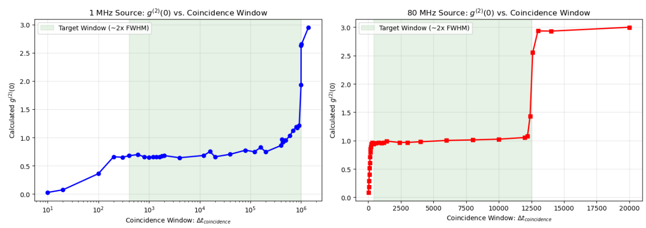

Figure 6: $g^{(2)}(0)$ measurement detailed in experiment 3.1.

These results were obtained in the laboratory session following the detailed steps below:

- **Hardware Setup:** Connect SPAD A, SPAD B, and laser sync to Swabian channels 1, 2, and 3 respectively. Align the laser to the SPAD active areas, insert the ND0.6 filter, and isolate the system from external light.

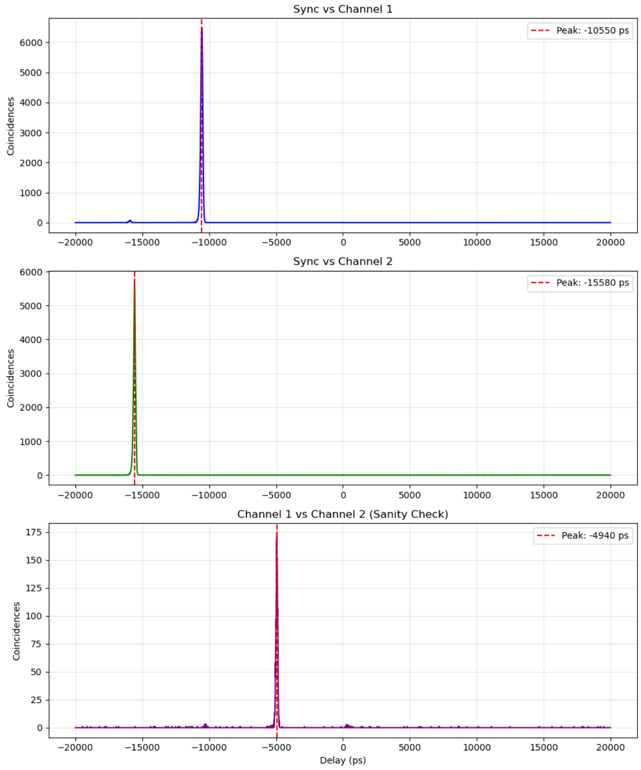

Figure 7: Differences in signal times. Calibration using `tt.Correlation` function.

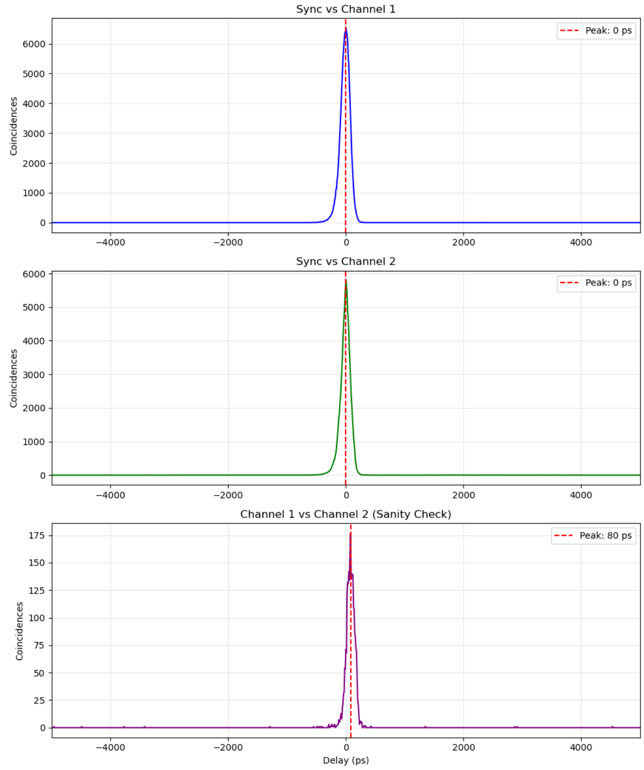

Figure 8: Calibrated signals to zero ps delay.

- **1 MHz Configuration:** Laser configured to 1 MHz (*Source: Int2*, *Rep Frequency: 1*) at 36.8% power. Measured optical power is 3 μW.

- **80 MHz Configuration:** Laser configured to 80 MHz (*Source: Int1*, *Rep Frequency: 1*) at 36.8% power. Measured optical power is 100 μW.

- **Count Rate Calibration:** Averaged over 0.5s using the `tt.Countrate` function. The active detections are summarized in Table 2.

- **Optical Path Delay Calibration:** Time differences between channels were extracted using the `tt.Correlation` function at 10 ps resolution. The peak delays are summarized in Table 3 (see Figure 7 and associated calibration plots).

- **Delay Application:** Delays were fed back into the Swabian hardware via `tt.setInputDelay(SPAD, delay)`. The motivation for this adjustment is to precisely align the SPAD signals to their respective laser sync signals. This ensures that the time tagger accurately counts the number of pulses, individual events, and coincidences corresponding to the exact same laser pulse (see Figure 8).

- **$g^{(2)}(0)$ Measurement:** The second-order coherence measurement was executed per the procedure in Subsection 3.1. Results for both frequencies are referenced in Figure 6.

Table 2: Average count rates (cps) over 0.5s

State	SPAD A cps	SPAD B cps	Laser cps
1 MHz (Laser On)	26,262.8	30,842.6	999,995.8
80 MHz (Laser On)	839,914.6	683,993.5	79,999,739.6

Table 3: Optical path delays via `tt.Correlation` (10 ps resolution)

Frequency	Sync→SPAD A	Sync→SPAD B	SPAD A→SPAD B
1 MHz Run	$-10,550$ ps	$-15,580$ ps	$-4,940$ ps
80 MHz Run	$1,970$ ps	$-3,050$ ps	$-4,930$ ps

After extracting the coincidences and calculating the zero-delay correlation, the expected and observed $g^{(2)}(0)$ values were analyzed. It is important to state that this specific measurement evaluates the macroscopic probability of detecting a pulse across two spatial channels, as opposed to the intrinsic quantum photon statistics of the pulsed laser source. For a high-power pulsed laser, the normalized second-order intensity correlation is expected to give $g^{(2)}(0) = 1$. This is due to the fact that in each pulse (whose FWHM is typically < 90 ps) there will be sufficient photons to trigger both SPAD detectors. While the 80 MHz run behaved as expected, the 1 MHz configuration—operating at a highly attenuated optical power of $1\,\mu$W—gives a low-valued correlation plateau of $g^{(2)}(0) \approx 0.6$, as shown in Figure 6.

This unexpected result for the 1 MHz source can possibly be due to electrical crosstalk within the SPAD array. At low photon fluxes, an avalanche in one pixel can cause an excess voltage drop across the shared electronics. If true, this blinds the neighboring SPAD during the brief < 90 ps laser pulse window, artificially reducing coincidence counts. If this hypothesis holds, increasing the incident optical power to drive the detectors into a saturated, multi-photon regime should overcome the blinding and recover the expected flat $g^{(2)}(0) = 1$ signal. This modified high-power condition is tested in the subsequent section.

3.1.2 Pulsed laser Measurement

In Figure 6, we did not obtain the expected value of $g^{(2)}(0) = 1$. We believe the discrepancy is a consequence of the low operational power (3 μW), where a low average number of photons per pulse leads to low single-photon detection probabilities and low count rates. This affects the statistical precision of the coincidence measurement against baseline timing fluctuations. To test the impact of statistical precision, a new measurement was conducted at 1 MHz with an increased optical power of 33.5 μW. An increase on the average photons per pulse, increases the corresponding detection probability, yielding a better signal-to-noise ratio without altering the intrinsic $g^{(2)}$ properties of the light source. Results are shown in Figure 9.

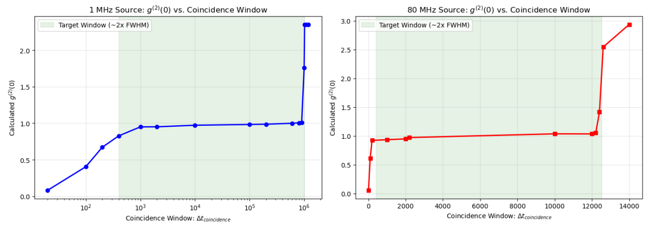

Figure 9: $g^{(2)}(0)$ measurement detailed in experiment 3.1.

The procedure is listed bellow:

- **Hardware Setup:** Connect SPAD A, SPAD B, and laser sync to Swabian channels 1, 2, and 3 respectively. Align the laser to the active areas and isolate the system from external light.
- **1 MHz Configuration (High Power):** Laser configured to 1 MHz (*Source: Int2*, *Rep Frequency: 1*) at 100% power. Measured optical power is 33.63 μW.
- **80 MHz Configuration:** Laser configured to 80 MHz (*Source: Int1*, *Rep Frequency: 1*) at 36.8% power. Measured optical power is 100 μW. An ND0.6 filter is applied to throttle the active counts into the $\approx 1\%$ detection domain.
- **Count Rate Calibration:** Averages extracted using `tt.Countrate` over 0.5s. Active detections for the multi-run 1 MHz measurements are summarized in Table 4.
- **Optical Path Delay Calibration:** Time differences extracted using `tt.Correlation` at 10 ps resolution. Peak delays for all runs are summarized in Table 5.
- **$g^{(2)}(0)$ Measurement:** Calculated delays were fed back into the Swabian hardware via `tt.setInputDelay()`. With the enhanced count rates providing sufficient statistical precision to resolve the true correlation curve, the measured $g^{(2)}(0)$ value successfully approaches 1

across the pulse duration, matching classical expectations.

Table 4: Average count rates (cps) over 0.5s (1 MHz Source)

State	SPAD A cps	SPAD B cps	Laser cps
1 MHz (Laser On)	665,354.0	584,594.0	999,996.0

Table 5: Optical path delays via `tt.Correlation` (10 ps resolution)

Frequency	Sync→SPAD A	Sync→SPAD B	SPAD A→SPAD B
1 MHz (Run 2)	$-10,140$ ps	$-15,160$ ps	$-4,910$ ps
1 MHz (Run 3)	$-10,190$ ps	$-15,200$ ps	$-4,930$ ps
80 MHz	$1,970$ ps	$-3,050$ ps	$-4,930$ ps

As a secondary result, the system jitter was extracted directly from the timing histograms shown in Figures 7 and 8. By evaluating the Full Width at Half Maximum (FWHM) of the cross-correlation peaks between the laser sync signal and each channel, the total instrument transit jitter values for SPAD A and SPAD B were determined to be 181.4 ps and 152.1 ps, respectively, as illustrated in Figure 10.

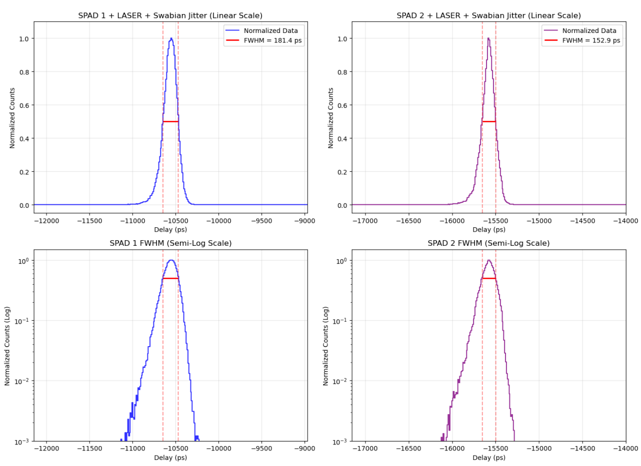

Figure 10: Optical setup jitter made up of Swabian, laser, and SPAD.

3.2 2nd Measurement: $g^{(2)}(\tau)$

The second set of measurements is designed to extract the second-order coherence function, $g^{(2)}(\tau)$. Note the distinction between $G^{(2)}(\tau)$, which represents the raw, unnormalized coincidence count histogram as a function of temporal delay, and $g^{(2)}(\tau)$, which is the normalized second-order intensity correlation. The procedure is as follows:

1. Feed the pulsed laser into the optical setup shown in Figure 1.
2. Select a laser repetition rate within the operational range (1 MHz to 80 MHz).
3. Build the raw coincidence histogram, $G^{(2)}(\tau)$.
4. Normalize the histogram by the average count of the time bins far from $\tau = 0$ to obtain $g^{(2)}(\tau)$.

For the pulsed laser, we expect to observe two distinct behaviors depending on the programmed bin width:

- If the bin width matches the period of the pulsed laser, each correlation peak is integrated into a single histogram bin, yielding an expected $g^{(2)}(\tau) = 1$.
- If the bin width is shorter than the period of the pulsed laser, we expect to resolve a comb structure corresponding to the individual laser pulses.

In contrast, for true thermal sources, the zero-delay correlation sits at $1 < g^{(2)}(0) \leq 2$ and decays to 1 over the coherence time. The temporal width of this decay is a direct measure of the coherence time [1].

3.2.1 Pulsed laser Measurements

Results are summarized in Figures 15 and 17. Each figure comprises three subplots illustrating the data processing pipeline: the raw coincidence histogram $G^{(2)}(\tau)$ (grey), the

[1] For true broadband thermal sources, this coherence time is fundamentally determined by the optical bandwidth of the emission.

correlation normalized via the Swabian API $g^{(2)}_{\text{swabian}}(\tau)$ (blue), and the manually normalized correlation $g^{(2)}_{\text{python}}(\tau)$ (red).

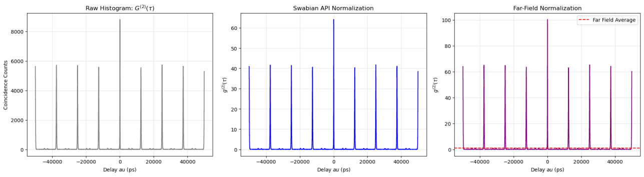

Figure 11: Cross-correlationfunction measurement for the laser operating at $80MHz$ with its respective period of $12,500ps$.

A similar result can be obtained for the $1MHz$ signal with its characteristic period shown in Figure 12.

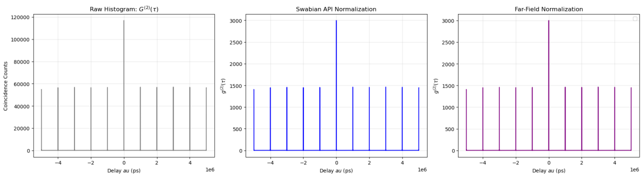

Figure 12: Cross-correlationfunction measurement for the laser operating at $1MHz$ with its respective period of $1,000,000$ps.

Figures 11 and 12 are indicators that our setup is measuring the intended correlations. Note they are not properly normalized $g^{(2)}$ because the normalization is taking into account the secondary peaks. However, when we tailor the `binwidth` of the correlation function to match the laser period, we observe a flat baseline (Figures 13 and 14).

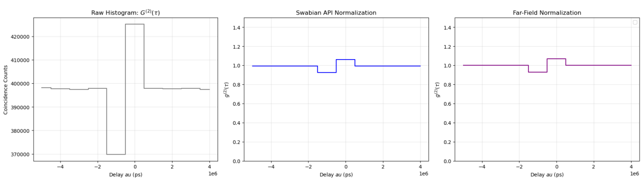

Figure 13: Cross-correlationfunction measurement.

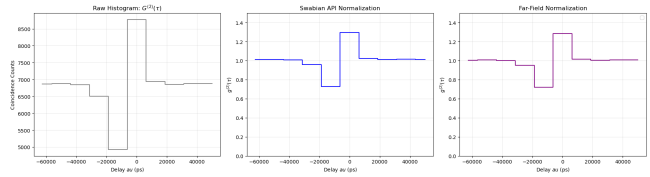

Figure 14: Cross-correlationfunction measurement.

Despite measuring the expected baseline in figures 13 and 14, they exhibit a severe quantization artifact: the digital Bin 0 overcounts, and consequently, Bin -1 significantly undercounts. This issue arises because the digital bin width is centered around 0 (i.e $\tau_{bin0} = [-1/2f_{laser}, +1/2f_{laser}]$ rather than $\tau_{bin0} = [0, +1/f_{laser}]$). To resolve this bin-edge quantization error, an artificial hardware offset equal to exactly half the bin width was introduced via the input delays. This offset physically shifts the photon arrival times such that the coincidence peak lands dead-center inside Bin 0, safely away from the edges. The corrected measurement, successfully eliminating the binning artifacts, is presented in Figure 15.

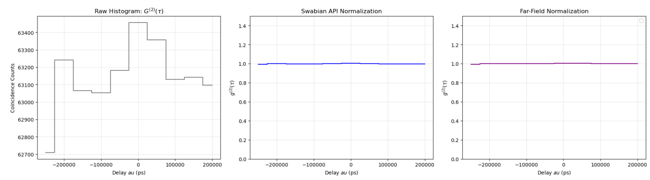

Figure 15: Cross-correlation function measurement.

While setting the correlation bin width equal to the laser's repetition period yields a flat $g^{(2)}(\tau) = 1.0$ response, this must not be misinterpreted as continuous Poissonian emission. Rather, it is a temporal integration artifact where each macroscopic bin perfectly envelopes exactly one discrete pulse.

3.2.2 Thermal Source Measurements

Next, we turn our attention to the thermal sources. We shine a thermal lamp into the SPAD and obtain its correlation function.

- Place a 2.3 ND filter over the SPAD.
- Turn on the thermal lamp or the laboratory light.

- Measure count rate: $\approx 1,000,000$ cps.
- Measure $G^{(2)}(\tau)$ and normalize to obtain $g^{(2)}(\tau)$.
- Repeat for increasing count rates.

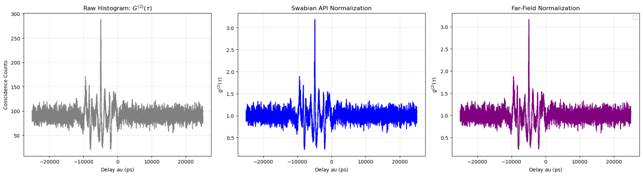

Figure 16: Cross-correlation function measurement for a thermal lamp.

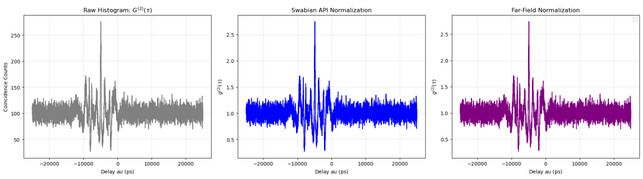

Figure 17: Cross-correlation function measurement for the laboratory light.

Notice the peak near $\tau = 0$ spanning approximately ± 10 ns in Figures 17 and 16. This does not represent true quantum thermal photon bunching. The picosecond resolution of the detectors and electronics is orders of magnitude longer than the femtosecond coherence time of these broadband sources, any actual thermal bunching peak is entirely time-averaged out. Instead, we argue that the central peak at zero delay is a hardware artifact present in SPAD arrays, driven by optical crosstalk [2]. However, the oscillations surrounding this central peak cannot be explained by crosstalk alone. These may be attributed to electronic timing effects, such as signal ringing or reflections in the readout circuitry, as well as potential normalization artifacts within the data processing window. Outside of this nanosecond hardware interference region, the correlation function correctly rests at a flat baseline of $g^{(2)}(\tau) = 1$, consistent with the expected temporal averaging of ultrafast thermal statistics.

[2] Hot-carrier luminescence between adjacent pixels triggering false coincidences.

3.2.3 CW Laser Source Measurements

Once the cross-correlation measurement pipeline was behaving as expected using the pulsed laser (section 3.2.1) and its temporal limits were established via the thermal sources (section 3.2.2), the $g^{(2)}(\tau)$ of a continuous-wave laser source was measured. In particular, we utilized the CW-laser 2 listed in section 3.3.

To ensure the SPAD array operated within a linear regime, we used the result from the autocorrelation measurement in figure 2. The value of $1/4.9ns \approx 200Mcps$ defines the absolute theoretical maximum count rate of the hardware. To prevent saturation and avoid statistical distortion from photon pile-up, the incident optical power was attenuated using neutral density filters below the 1% regime. The results shown in figure 18 where obtained at a rate of 1 Mcps. Under low-flux conditions, the pure phase-coherent state of the CW laser successfully yielded the expected flat correlation baseline of $g^{(2)}(\tau) = 1$ across a $10\mu s$ window.

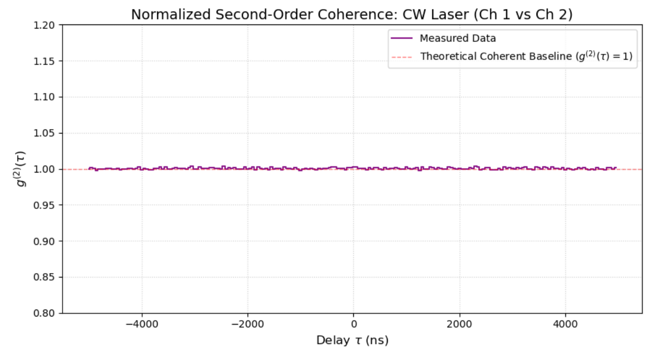

Figure 18: Cross-correlation function measurement for the continuous wave laser.

3.3 3rd Measurement: Diffuse Correlation Spectroscopy

As seen in the second measurement, true thermal light decorrelates in femtoseconds, which is too fast for the SPAD dead time to measure. To address this, we utilized a pseudo-thermal light source. The experiment shines the 785 nm continuous-wave (CW) laser through a rotating diffuser to generate a dynamic speckle pattern. The formula for the average speckle diameter $S \approx 1.22\lambda z_{det}/D$, where z_{det} is the distance from the diffuser to the detector plane, yields an estimated speckle diameter of 1.58 μm for our particular setup.[3]. By changing the motor speed, this moving diffuser creates macroscopic intensity fluctuations in the millisecond regime, allowing our Time Tagger to capture the decorrelation dynamics.

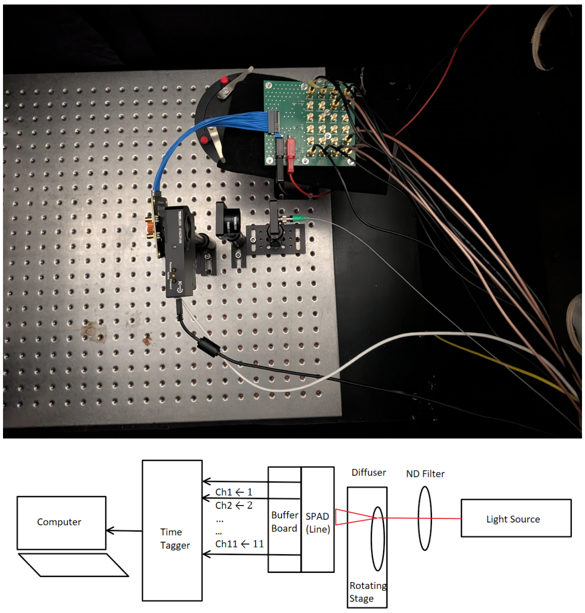

Figure 19: Dynamic speckle setup measurement. CW laser incident onto the rotating diffuser with the SPAD detector array positioned behind it (left).

The measurement procedure was as follows:

1. A 785 nm CW laser was aligned to hit the right edge of a rotating diffuser.

2. The line sensor SPAD array was placed at a distance $z_{det} = 37.5$ mm behind the diffuser to detect the speckles.

3. To avoid hardware buffer overflow from high count rates, we lowered the laser intensity to a safe kilocounts-per-second (kcps) level.

4. To recover the signal-to-noise ratio (SNR), we simultaneously measured the autocorrelation of 11 separate pixels to average them together.

5. The measurement was repeated for different diffuser angular velocities (5, 10, 15, and 20 deg/s).

Results

To process the data, each pixel's measurement was normalized by dividing it by its flat baseline. Figure 20 shows the 11 individual pixels (faded) and their combined average (black). Averaging the pixels gives a much smoother exponential decay curve, allowing us to accurately fit the data and extract the decorrelation time (τ_c). The data was fitted using the standard intensity correlation model:

$$g^{(2)}(\tau) = 1 + \beta \exp\left(-2\frac{\tau}{\tau_c}\right)$$

where β is the spatial coherence factor and τ_c is the characteristic decorrelation time.

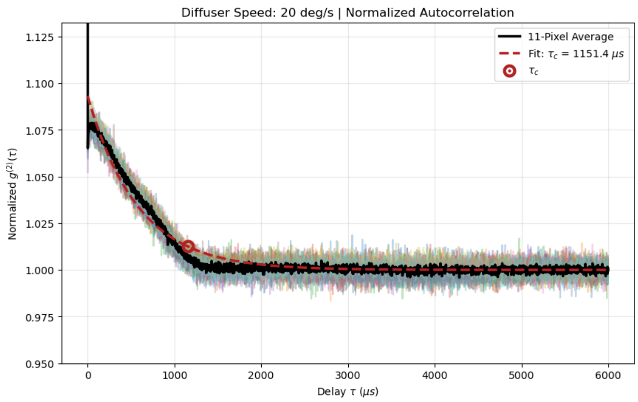

Figure 20: Normalized autocorrelation measurement for the rotating diffuser. The black line is the average of 11 pixels, which is fitted with an exponential curve (red dashed line) to extract τ_c.

Figure 21 demonstrates why using the 11-pixel array brings an advantage. As we average more pixels (N), the measured decorrelation time remains stable, but the baseline noise drops following the theoretical $1/\sqrt{N}$ rule.

Finally, Figure 22 plots the extracted τ_c against the different motor speeds. As the diffuser spins faster, the speckles move past the detector faster, causing the decorrelation time to drop. This inverse relationship is consistent with this proof-of-principle demon-

[3]Assuming a uniform circular scattering spot, an operating wavelength of 785 nm, and perfect laser alignment spanning exactly half of the diffuser's diameter (i.e an illumination spot size of $D = 22.7$ mm).

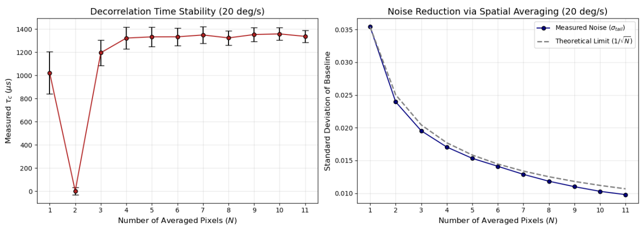

Figure 21: Effect of averaging multiple pixels. The measured τ_c remains stable (Left), while the noise drops with $1/\sqrt{N}$ (Right).

stration of Diffuse Correlation Spectroscopy (DCS) using a the pseudo-thermal source.

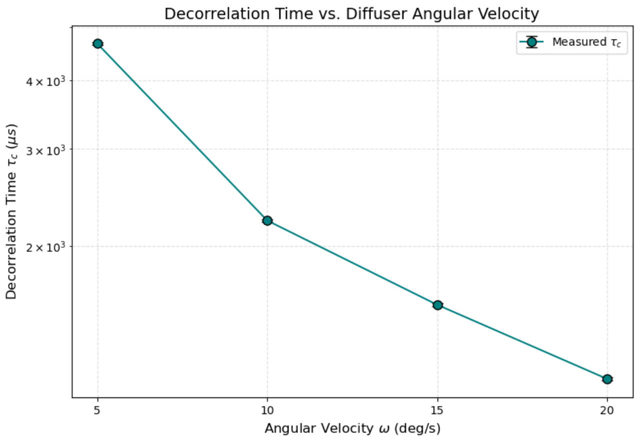

Figure 22: Decorrelation time (τ_c) measured at different diffuser speeds. As angular velocity increases, τ_c proportionally decreases.

Conclusion

This work demonstrated the use of single-photon avalanche diodes (SPADs) and time-correlated single-photon counting (TCSPC) to measure the second-order correlation function, $g^{(2)}(\tau)$, of various light sources. By characterizing the detector hardware, the fundamental limitations of the measurements were established, and the statistical signatures of coherent lasers and super-Poissonian sources were obtained.

An experimental data-processing pipeline was successfully implemented and validated for cross-correlation measurements. For pulsed laser illumination, the zero-delay cross-correlation, $g^{(2)}(0)$, was successfully measured by varying the coincidence criterion and under both high and low count-rate conditions, demonstrating the critical impact of statistical precision. Furthermore, mapping the full temporal correlation function, $g^{(2)}(\tau)$, successfully resolved the characteristic comb structure of the pulsed emission. By adjusting the correlation bin width to match the laser period, the expected flat integration baseline was recovered, validating the pipeline's capability to accurately capture and process cross-correlation dynamics.

The measurement of a non-coherent broadband source (fluorescent lab light) highlighted the temporal limits of the detector architecture. The femtosecond decorrelation times of such broad-bandwidth sources were shown to be entirely averaged out by the finite timing resolution of the SPADs and readout electronics. The residual signal observed at zero delay was attributed to electronic timing effects and optical crosstalk between adjacent pixels, rather than true quantum photon bunching.

Finally, a rotating diffuser was implemented to generate dynamic speckle patterns, providing a proof-of-principle demonstration of Diffuse Correlation Spectroscopy (DCS). Scattering a continuous-wave laser through this dynamic medium generated macroscopic speckle intensity fluctuations in the millisecond regime. This allowed the TCSPC hardware to accurately capture the exponential decorrelation of the dynamic speckle pattern. Furthermore, using the linear SPAD array through multi-pixel spatial averaging proved effective in reducing baseline measurement noise in accordance with the theoretical $1/\sqrt{N}$ scaling. Ultimately, this methodology captured the dynamic photon statistics, accounted for detector artifacts, and linked the extracted decorrelation times to the kinematic properties of the macroscopic target.

Appendix: Equipment

Lasers and Light Sources

- **Pulsed LASER:** PicoQuant PDL Series
  - **PicoQuant Laser Diode Head**
  - **Wavelength:** 640 nm
  - **Repetition Rate (f_{rep}):** Tunable frequencies (operated between 1 MHz and 80 MHz).
  - **FWHM of pulse:** < 90 ps
- **Continuous wave (CW) laser 1**
  - **Thorlabs S1FC660 (Fiber-coupled)**
  - **Wavelength:** 660 nm
  - **Power:** Tunable from 0 mW to 19.8 mW
  - **Application:** Used for the Poissonian Statistics Verification.
- **Continuous wave (CW) laser 2**
  - **Thorlabs CLD1015:** Diode Controller.
  - **Thorlabs DBR785P:** Diode head.
  - **Wavelength:** 785 nm
  - **Power:** Tunable diode current (between 62.7 mA and 65.5 mA)
  - **Application:** Used for the Pseudothermal Light Simulator.
- **Pseudothermal light simulator**
  - **Light Source:** CW laser 2
  - **Rotating Stage:** Thorlabs K10CR1/M
  - **Motor Speed:** Up to $20°/s$
  - **Diffuser:** Glass diffuser

Timing Electronics

- **Swabian Instruments Time Tagger**
  - Model: Ultra, USB 3.0
  - Max Data Transfer Rate: 90 MTags/s
  - Max Input Frequency: 475 MHz
  - Dead Time: 2.1 ns
  - Base Jitter: 42 ps RMS (100 ps FWHM)
  - Trigger Level Range: -2.5 V to +2.5 V

Detectors and Optical Components

- **2 Pixel SPAD Module**
  - **Model:** NV02TPX
  - **Array format:** 2 x 1 pixels.
  - **Pixel Pitch:** $28\mu m$
  - **Dark count rate:** 10 cps
  - **Quenching:** passive with active recharge
- **Linear SPAD Array Module**
  - **Model:** Novoviz Small Line Sensor
  - **Array format:** 12 x 2 pixels.
  - **Pixel Pitch:** $125\mu m$
  - **Quenching:** passive with active recharge
- **Illumination Diffuser**
  - Glass diffuser
  - N-BK7 Ground Glass Diffusers
- **Neutral Density Filters**
  - **Absorptive ND glass filters**
  - **OD:** 0.6 and 2.3
  - **Wavelength Range:** from 400 nm to 700 nm
  - Broadband AR coating operating
- **Fiber/Free Space Coupler:**
  - **Model:** Thorlabs SM1 FC2
  - Fiber Optic cable connector: Single Mode FC/APC

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